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Digital twins in healthcare: a review of AI-powered practical applications across health domains

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Abstract

This review examines the evolving role of digital twins (DTs) in healthcare and how artificial intelligence (AI) is shaping personalized medicine across various medical fields. Digital twins are virtual models that mirror individual patient profiles, making it possible to customize treatments and predict health outcomes more accurately. Through a refined selection process, we have identified 17 distinct applications of this technology in the past four years, each offering significant contributions to AI-driven healthcare innovation. This review highlights the progress of AI-powered digital twins in areas such as heart health, diabetic care, mental wellness, respiratory health, and stress management. To support reader understanding and accessibility, we present intuitive visuals that break down complex processes, aiming to give a clear view of AI's expanding potential to reshape healthcare toward more proactive and patient-specific outcomes.

Keywords: Digital twins, Healthcare, Personalized medicine, AI-driven digital twin applications, Generative models for digital twins, LLMs for digital twins, Medical digital twins, Virtual organs, Diabetes management, Stress management, Digital patient avatars, Virtual patient modeling, Heart digital twin, Mental health, Patient trajectory forecasting, Virtual clinical trials

Introduction

Within the broad landscape of modern medicine, the quest to tailor treatments to the unique needs of each patient has never been more crucial [1, 2]. As the complexity of diseases and the volume of medical data expand [3, 4], innovative approaches are essential to navigate this intricate landscape. Picture a world where every patient's journey through illness is not just a series of treatments and outcomes but a dynamic, interactive simulation that mirrors their individual experience. This is not the stuff of science fiction but the promise of digital twins in healthcare. By creating virtual replicas of patients' digital twins, scientists and doctors can explore and predict the effects of various treatments in a controlled, risk-free environment.

As cancer rates soar globally, with millions of new cases each year [5, 6], the pressure to find more effective, personalized treatment options intensifies. At the same time, the emergence of new diseases, the rise of chronic conditions, and the ongoing threat

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of new viruses further complicate the healthcare landscape. The COVID-19 pandemic, in particular, has underscored the critical need for resilient health systems capable of addressing these multifaceted challenges [7]. Traditional methods, while foundational, often fall short in capturing the nuanced and evolving nature of diseases such as cancer. Enter the digital twin, a sophisticated tool that allows clinicians to simulate and analyze a patient's potential responses to different therapies in real-time. The idea of a digital twin in healthcare extends beyond mere simulation; it involves a comprehensive digital representation of an individual's biological and medical status [8]. By integrating vast amounts of data ranging from clinical records, medical images to genomic sequences, digital twins offer a way to personalize treatment and improve outcomes. This approach not only promises to enhance the precision of cancer care but also to reduce the overall cost and complexity of treatment [9]. In this evolving field, digital twins are poised to transform how we approach healthcare. By leveraging advanced technologies such as artificial intelligence (AI) and machine learning (ML), these virtual models provide a deeper understanding of each patient's unique disease profile and treatment response. As we move forward, the potential to revolutionize patient care and achieve unprecedented levels of personalization in treatment becomes ever more attainable.

Overview

The roots of digital twins lie in the world of engineering, where the idea first emerged [10]. Engineers used digital models to simulate physical objects such as airplane engines or industrial machinery, allowing for real-time monitoring and predictive maintenance [11]. This approach was revolutionary, allowing issues to be detected before they occurred. At the core of a digital twin is a dynamic feedback loop that connects a physical entity such as a patient, organ, or physiological process with a digital counterpart that continuously evolves to reflect the real-world state of that entity. This connection is established through bi-directional data flow, allowing the digital twin to ingest real-time sensor data, clinical records, imaging, and lab results. And, in turn, simulate scenarios, forecast outcomes, and recommend interventions. As shown in Fig. 1 The underlying architecture typically consists of three main components: (1) the physical system, (2) a virtual model that simulates the physical counterpart using statistical, mechanistic, or AI-based methods, and (3) a data connection that synchronizes both. Over time, the concept of digital twins began to gain traction in healthcare, driven by advances in data science and computational power. Researchers and practitioners recognized the potential to apply this technology to human health, envisioning digital replicas of patients that could dynamically reflect their biological and physiological states [12]. The early stages of this leap were marked by the development of more sophisticated models and simulations. At this point, healthcare digital twins were largely theoretical, relying on existing technologies such as electronic health records and genetic sequencing. The goal was to create virtual models that could provide personalized insights and predict health outcomes with greater accuracy. Building on the engineering success of digital twins, the concept matured and found widespread utility across various sectors [13], particularly in aviation. In the 1970 s, NASA used mirrored systems for spacecraft monitoring and simulation [14]. An example is the Apollo program, where an identical space capsule was kept on Earth to simulate the behavior of the capsule in space [15]. This physical replica,

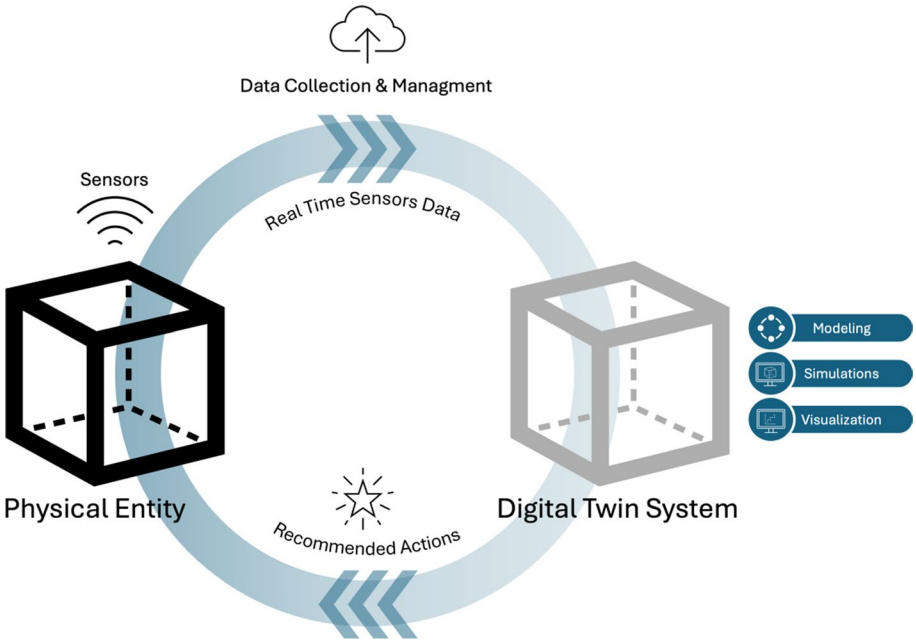


Fig. 1 Conceptual representation of the standard architecture of a digital twin system, illustrating the integration of real-time data, computational modeling, visualization, and recommended action

while useful, faced limitations due to constraints in data interchange and synchronization. The concept of digital twins continued to evolve, leading to NASA's definition of digital twins in 2010 [16]. This definition underscored their potential to simulate and mimic the entire lifecycle of complex systems. In the context of autonomous systems, digital twins offer a powerful framework for simulating the behavior of vehicles, predicting potential failures, and optimizing system performance. By creating a virtual counterpart of an autonomous vehicle [17, 18], engineers can test various driving scenarios, ensure safety compliance, and refine decision-making algorithms in real time. This integration plays a pivotal role in improving the resiliency and security of self-driving vehicles, ensuring safer transportation environments for both passengers and pedestrians [19]. As data-driven vehicles continue to evolve, digital twins will be indispensable in enhancing the design, operation, and safety of autonomous vehicles in smart transportation networks [20]. Another critical application of digital twins (DTs) is in automated pharmaceutical laboratories [21]. In Quality Control (QC) labs, the complexity of workflows and simultaneous testing of drug samples necessitates a high degree of efficiency and precision. By integrating digital twins into Cyber-Physical Production Systems (CPPS), pharmaceutical labs can simulate and optimize their workflows, reducing human error and improving overall productivity [22]. The relentless advancement of artificial intelligence, Big Data, and Internet of Things (IoT) technologies has further propelled the evolution and adoption of digital twins across various domains. In healthcare, this integration has led to the development of more detailed and dynamic virtual models of patients. By leveraging vast amounts of data, including genetic information and real-time health monitoring, digital twins in healthcare can provide highly personalized insights and predict health outcomes with greater accuracy. The transition from theoretical models to practical applications has been noticed by significant progress

in developing digital representations of individual patients. These models promise to enhance personalized medicine by enabling more tailored and effective medical interventions. As the technology continues to advance, the potential applications of digital twins in healthcare are expanding, offering new possibilities for preventative care, human enhancement, and optimized treatment strategies. The history of digital twins in healthcare reflects a remarkable journey from engineering origins to their current role in personalized medicine. From their early applications in engineering to their evolution and widespread use in various sectors, digital twins have transformed how we understand and manage complex systems. As the technology continues to advance, it will be crucial to address the associated ethical and social challenges, ensuring that the benefits of digital twins are realized while mitigating potential risks.

The primary goal of this review is to explore the current landscape of digital twin applications in healthcare, with a specific focus on how AI-driven technologies are transforming personalized medicine across various medical fields. We review real-world implementations where patient data has been utilized effectively.

We apply strict criteria for inclusion: each study must provide a concrete example of DT technology applied in practice, utilizing specific datasets to achieve measurable outcomes. By concentrating on these cases, this review aims to identify emerging trends in healthcare DT applications, highlighting clear results and the practical potential of DTs in advancing personalized medicine. Through this approach, we seek to provide insights into how DTs are evolving from theoretical concepts into tangible tools for enhancing healthcare.

Methodology

In this section, we describe the process of selecting and extracting articles to address our research objectives.

Figure 2 shows the flow diagram used to outline the process of identifying and selecting studies. We started by searching databases with a refined query to gather articles. After removing duplicates, we screened the remaining articles by checking the titles and abstracts to see if they were relevant. Next, we reviewed the full text of each selected article, excluding those that did not fit our research question. This process gave us a final set of 17 studies to analyze.

Article collection process

To identify the main trends in the use of Digital Twins (DTs) in healthcare, we conducted a targeted search of peer-reviewed articles from two prominent academic databases, Web of Science [23] and Scopus [24]. The search strategy was designed and refined to ensure it captures studies most relevant to our specific research focus. Table 1 outlines the specific optimized queries we applied to both databases. We also applied consistent filters across both databases: only articles published in English were included, and all peer-reviewed formats (including journal articles, conference papers, and reviews) were considered. These filters were applied in conjunction with the search queries detailed in Table 1 to ensure relevance and methodological consistency. The resulting articles span the period from 2019 to 2024, reflecting the most recent developments in the field. In total, we retrieved 206 articles. This number includes 44 from Web of Science, 77 from

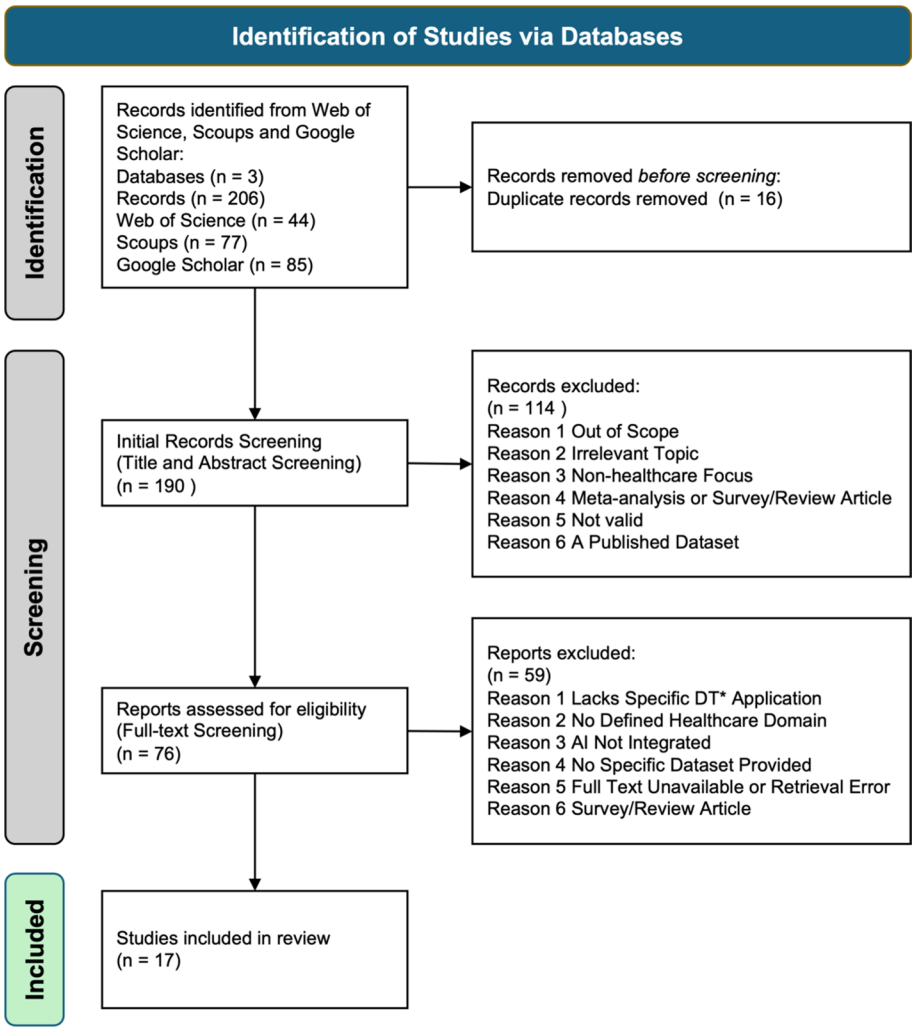


Fig. 2 The process of article identification, screening, exclusions, and final inclusion from database searches

Table 1 Queries used for extracting articles from Scopus and Web of Science databases

Query source	Query term	Number of results
Web of Science	((((((ALL=(Digital twins)) AND TS=(Healthcare)) AND TS=(AI OR "artificial intelligence" OR "generative AI" OR "large language models")) AND LA=(English)) NOT ALL=(blockchain)) NOT ALL=(metaverse))	44
Scopus	(TITLE ("Digital twins") AND TITLE-ABS-KEY (healthcare) AND TITLE-ABS-KEY ("AI" OR "artificial intelligence" OR "generative AI" OR "large language models") AND NOT TITLE-ABS-KEY (blockchain) AND NOT TITLE-ABS-KEY (metaverse) AND NOT TITLE-ABS-KEY ("Internet of Things" OR iot)) AND (LIMIT-TO (LANGUAGE, "English"))	77

Scopus, and 85 articles that were manually selected from Google Scholar. These manually chosen studies were included because they were identified as highly relevant to our research focus but were not captured by the database queries. After compiling the initial

set of 206 articles, we systematically removed duplicates, resulting in a final count of 190 unique articles.

Initial screening process

We began with an initial screening of titles and abstracts to assess each article's relevance to our research. This step helped us ensure that selected articles focused on our main question and core themes, including digital twins, healthcare, and AI. Articles were excluded if they did not meet the following criteria:

- Out of Scope: Studies focusing on digital twins in unrelated fields like cybersecurity, smart homes, cities, or communication systems were removed.
- Irrelevant Topic: Articles that did not clearly address Digital Twins or lacked sufficient application within healthcare were excluded.
- Non-healthcare Focus: Studies not related to healthcare were removed from further consideration.
- Meta-analyses and Reviews: Systematic reviews, surveys, and meta-analyses were excluded.
- Unavailable Articles: We excluded papers where the abstract or full text was inaccessible.

After this initial screening process, a total of 114 articles were excluded, leaving 76 articles that moved on to the full-text screening phase.

Full-text screening

For the remaining articles, we carefully conducted a thorough full-text review to ensure alignment with our research purposes. In this process, we aimed to guarantee the relevance and quality of the selected articles by addressing specific research questions that informed our evaluation criteria. The first criterion assessed whether the articles presented a direct application of digital twins in healthcare or if they were predominantly theoretical. Articles demonstrating practical applications were included in our review, while those that lacked real-world applicability were excluded. We also identified the specific healthcare domains in which digital twins were applied, such as chronic diseases, cancer, diabetes, cardiovascular diseases, and stress management.

Another essential aspect of our screening involved evaluating whether the applications of digital twins incorporated artificial intelligence, machine learning, or any form of intelligent algorithm or tool. Articles that clearly integrated AI into their methodologies were included, whereas those lacking this integration were excluded. We catalogued the types of AI models employed within the articles, such as machine learning algorithms, generative AI techniques, large language models (LLMs) and graph theory, providing insight into the diverse applications of AI in the field.

Data utilization played a significant role in our screening process. We prioritized articles that explicitly defined and utilized datasets, including them in our review. Conversely, articles that did not clarify their datasets, mentioned only general DT development, or could not be retrieved were excluded. We also assessed the role of data integration in enabling practical applications.

This systematic approach Helped retain only the articles that directly answered our research question and met our specific criteria for analysis. This thorough screening is crucial to our review, enabling us to gather valuable insights on the role of digital twins and AI in Healthcare. After completing the full-text review, we identified 17 articles that will be examined to address our research questions. Figure 3 shows the steps taken during the full-text screening process.

Digital twins applications across multiple medical domains

In this section, we organized our review into focused subsections, grouping related studies under clear headings. For example, applications of digital twins in cardiac care are collectively presented under the subsection Simulating the Heart: digital twins for Precision Cardiac Care, which includes several studies on Heart-focused digital twin

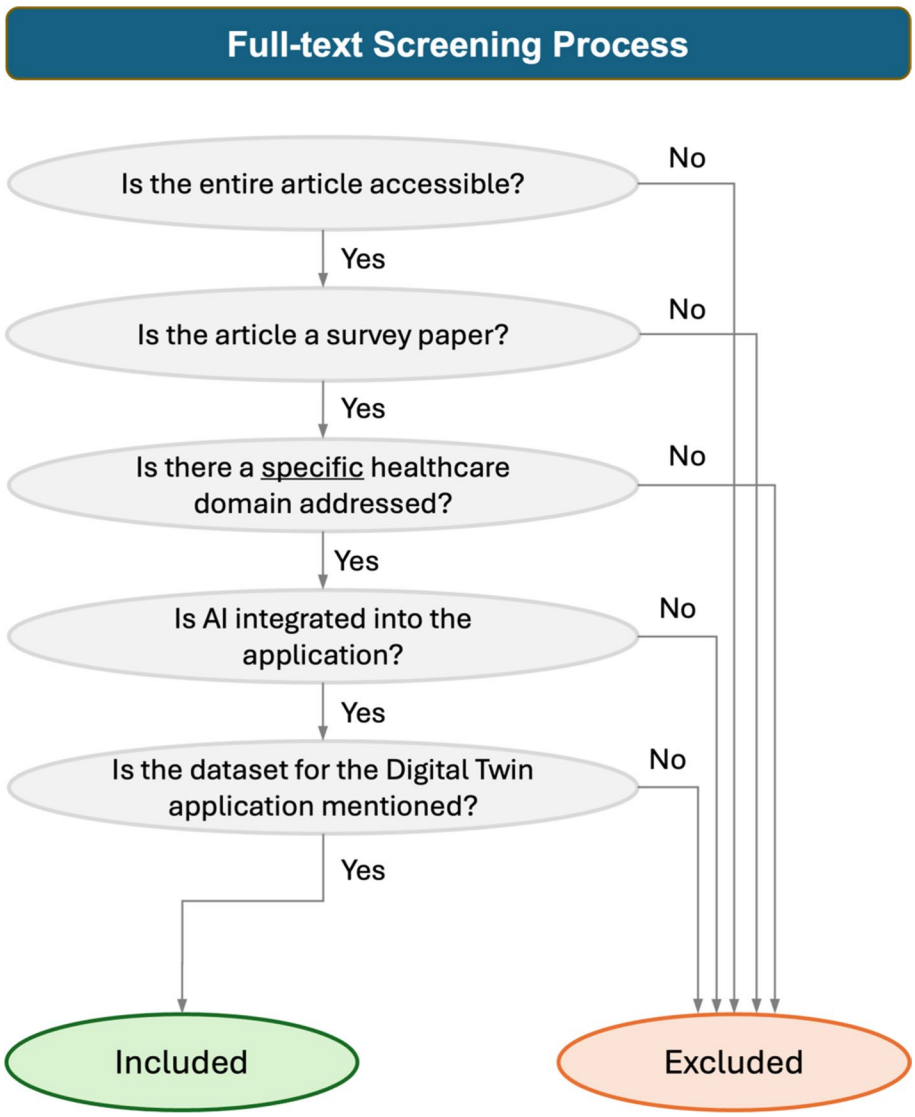


Fig. 3 Visual guide illustrating the full-text screening process, including rules for article inclusion and exclusion

technologies. Table 1 in Supplementary data brings together all the studies reviewed in this paper, listing the year, authors, application domain, and outcomes, providing a quick overview of focus and impact of each study on healthcare.

Table 2 lists the available performance metrics for some studies; metrics vary between papers and are omitted when not reported. For complete results, please refer to the original publications.

Simulating the heart: digital twins for precision cardiac care

One such example proposed by Rouhollahi et al., which addressed the challenge of aortic stenosis (AS) [37], a common heart valve disease predominantly affecting the elderly, where the narrowing of the aortic valve due to calcification complicates treatment planning [25]. Particularly, procedures like transcatheter aortic valve replacement (TAVR) [38] require precise anatomical understanding. Traditional methods of generating patient-specific digital replicas for surgical planning are time-consuming, complex and require expertise. To overcome this, the study introduces CardioVision, an AI-driven tool designed to automatically generate digital twins of AS patients using deep learning techniques. CardioVision automates computed tomography (CT) image segmentation and reconstructs 3D models of the aortic valve, offering a highly detailed view of the valve and surrounding structures. These digital twins allow clinicians to simulate surgical procedures like TAVR, predict complications, and personalize prosthetic selection,

Table 2 Summary of reviewed studies and their reported performance metrics

Authors	Application	Performance metric(s) reported
Rouhollahi et al. [25]	Cardiac care	IoU: 95.6%, Dice score: 97.8%
Dervisoglu et al. [26]	Cardiac care	Model: LR + Binning + chi2; accuracy: 0.86; recall: 0.98; F-score: 0.92. Detailed results in Table 1 of the original work
Shamanna et al. [27]	Diabetes	Diabetes-related tests performed. Detailed results in Tables 1 and 2 of the original study
Abirami et al. [28]	Mental health	*DS1:* accuracy: 97.95%, F1-score: 0.98, MCC: 0.93675; *DS2:* accuracy: 91.48%, F1-score: 0.91, MCC: 0.79816. Detailed results in Table 3 of original work
Abilkaiyrkyzy et al. [29]	Mental health	Accuracy: 69%, Acceptability and usability score: 84.75%. Full results in Tables 4–6 of the original work
Sarp et al. [30]	Chronic wound management	MSE: 25.84
Avanzato et al. [31]	Lung health	Accuracy: 96.8%; precision: 92%; recall: 97%; F1-score: 94%
Kolekar et al. [32]	Lung cancer	BSC: C-index = 0.97; LCSA: C-index = 0.82; Rapid response—F1: 0.65, sensitivity: 0.77, AUROC: 0.84, AUPRC: 0.84
Kumi et al. [33]	Stress management	MAE: 3.3988; MAPE: 4.6020; RMSE: 4.4702; R^2: 0.6321
Wang et al. [34]	Clinical trials	Pearson r: up to 1.0; AUROC: 0.821; Presence disclosure sensitivity: 20%; Attribute disclosure sensitivity:< 0.25; NNAA score: 0.271
Marakov et al. [35]	Patient Health	*NSCLC:* MAE = 0.55 ± 0.04 (↑3.4%), R^2 = 0.98; *ICU:* MAE = 0.59 ± 0.03 (↑1.3%), R^2 = 0.99
Moore et al. [36]	Oncology	AUROC: 0.864 (95% CI: 0.857–0.872) with SynTwin; baseline AUROC: 0.791 (95% CI: 0.781–0.800) with real data

ultimately optimizing patient-specific treatment strategies. The digital twin development uses the U-Net deep learning architecture to segment CT scans and reconstruct patient-specific geometries, focusing on calcification a critical factor in AS progression. By utilizing 35 CT scans, the input data were raw images, while the output was a highly accurate 3D digital twin. This model aids in visualizing calcification distribution, which directly impacts valve function and the success of interventions like TAVR. The evaluation of CardioVision showed a high accuracy rate, with metrics such as an Intersection over Union (IoU) [39] of 95.6% and a Dice Score of 97.8%. These reconstructions enhance pre-surgical planning and prosthetic device fitting, leading to personalized care and reduced complications. Lwin, T.C. et al. shifted the focus from adult heart disease to fetal heart monitoring, leveraging digital twin technology to improve fetal heart rate (FHR) monitoring [40] during labor [41]. Fetal heart rate variability (FHRV) [42] is a crucial indicator of fetal well-being, helping predict complications and assess potential long-term risks, such as non-communicable diseases (NCDs). The team created a digital twin to simulate the fetal heart's real-time behavior during labor, enabling better monitoring and prediction of fetal outcomes. To enhance the accuracy of this digital twin, the system integrates three entropy measures [43]: Shannon, Approximate, and Sample Entropy to assess the complexity of the fetal Heart rate. These measures Helped create a more detailed and predictive virtual model of fetal heart rate dynamics. The study analyzed 585 fetal electrocardiogram (ECG) recordings collected via scalp electrodes during labor in Japan. By processing this data, the digital twin provided real-time simulations of fetal health, focusing on key indicators such as umbilical cord blood gas parameters [44] (bicarbonate and base excess). These parameters are critical for ensuring fetal well-being and informing medical interventions. By accurately correlating entropy measures with blood gas data, the digital twin system was able to anticipate potential complications, providing early warnings to clinicians. This proactive system highlights the potential of digital twins in enhancing prenatal care through personalized monitoring. Dervisoglu, H. et al. tackled the critical need for continuous monitoring and timely intervention in cardiovascular diseases (CVD), a leading global cause of death [26]. The study compared two digital twin frameworks, cloud-based and edge-based, to evaluate their effectiveness in real-time prediction and monitoring of heart disease in patients. Both frameworks were designed to predict heart disease using ECG data and to send real-time alerts when abnormalities were detected. The digital twin technology allowed for real-time virtual representation of patient heart health, enabling early detection and timely interventions. The cloud-based framework leveraged centralized data storage and historical analysis, while the edge-based framework used local processing for real-time analysis, reducing latency. The study utilized the PTB Diagnostic ECG Database [45], which includes 549 ECG records from 290 patients (healthy and cardiac). Each record contained 15 ECG signals, processed through both cloud and edge frameworks. The edge-based framework showed lower latency (10.6 ms) compared to the cloud system (18.8 ms), making it more suitable for real-time cardiac monitoring. In terms of accuracy, both systems performed well, with the Logistic Regression [46] model achieving 86% accuracy and 98% recall. The Gradient Boosted Tree model [47] demonstrated the highest precision at 92%. These findings suggest that edge-based systems are better suited for real-time monitoring, while cloud-based systems offer greater scalability for larger datasets. Both

frameworks hold significant potential for improving early detection and management of heart disease. We noticed that most of the studies rely on relatively small datasets, raising concerns about their scalability and performance across diverse patient populations. Addressing this challenge will be crucial to ensuring that digital twin technology can be effectively and reliably implemented in heart disease management on a larger scale (Fig. 4).

Diabetes management through digital twin technology

One of the most significant applications of digital twin technology is its use in diabetes management [48, 49]. Diabetes is a chronic condition that requires constant monitoring and precise adjustments in treatment based on individual patient responses to factors such as diet, physical activity, and insulin levels. Traditional methods often fall short in capturing the complexity and variability of each patient's condition, leading to challenges in maintaining optimal blood glucose levels. In this section, we provide

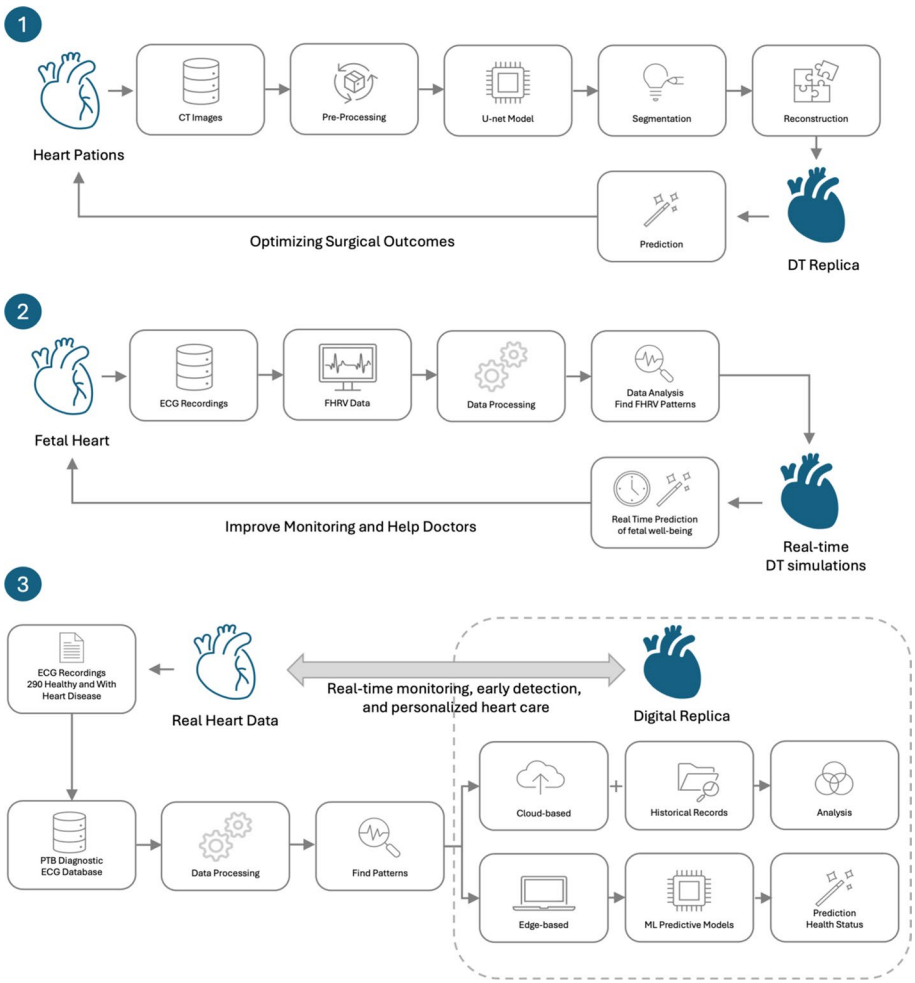


Fig. 4 Overview of digital twin applications for heart-related studies, illustrating the workflows for optimizing surgical outcomes in aortic stenosis, real-time fetal heart rate (FHR) monitoring, and predictive heart disease models using cloud-based and edge-based DT frameworks. Each section highlights the key steps from data acquisition to prediction, demonstrating the role of digital twins in enhancing heart health management

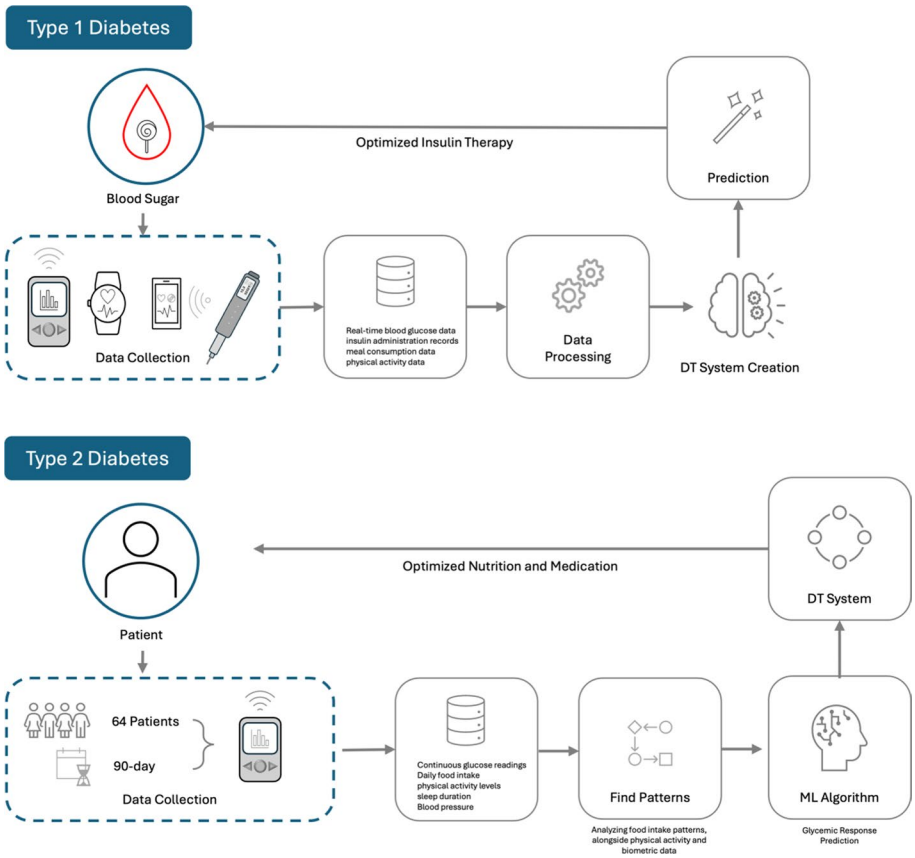


Fig. 5 Illustration of digital twin frameworks for Type 1 and Type 2 diabetes management. For Type 1 diabetes, the system integrates real-time glucose monitoring, insulin administration records, and physical activity data to optimise insulin therapy through continuous data processing and prediction. For Type 2 diabetes, a machine learning-based digital twin analyzes glucose readings, food intake, and biometric data from 64 patients over 90 days to identify patterns, enabling precision nutrition and optimized medication management. The figure highlights how digital twin technology tailors personalized therapy for both conditions

a review of different applications. Figure 5 represents the application of digital twin technology in managing both Type 1 and Type 2 diabetes, as demonstrated in two studies. These studies highlight how digital twins are being leveraged to personalize treatment and improve outcomes by providing real-time, data-driven insights for insulin therapy and nutrition optimization.

Cappon et al. introduced the TWIN system, a digital twin-based clinical decision support system (DSS) designed to enhance the management of Type 1 diabetes (T1D) in pediatric patients [50]. Managing T1D [51], poses significant challenges due to the constant need to balance evolving physiological needs and maintain stable blood glucose levels. The standard approach of multiple daily insulin injections (MDI) requires precise dosage adjustments based on a variety of factors, such as diet, physical activity, and patient behavior, making diabetes management labor-intensive and error-prone. To address these challenges, the TWIN system integrates continuous glucose monitoring (CGM) devices, smart insulin pens, and physical activity trackers to automate the collection of real-time patient data. The digital twin technology creates a

virtual model, allowing detailed simulations of how their body responds to different insulin dosages. This personalized approach enables healthcare providers to make more accurate, data-driven decisions for insulin adjustments, improving treatment efficacy.

At the core of TWIN's digital twin generation process is the ReplayBG algorithm, a nonlinear model that simulates glucose concentration responses based on real-time data. By leveraging Bayesian techniques, the system estimates physiological parameters unique to each patient and continuously runs simulations to optimize insulin therapy, both for shortacting and long acting insulin doses. Additionally, the system's AI-driven suggestion module provides healthcare providers with dosage recommendations, explaining the reasoning behind them through a large language model similar to Generative Pre-trained Transformer (GPT). The TWIN system significantly reduces manual data entry for patients and caregivers while ensuring a high level of accuracy in therapy recommendations. Real world data from CGM devices, insulin administration logs, and physical activity trackers are fully integrated into the system to deliver optimized insulin therapy parameters. The system's outputs include personalized adjustments for carbohydrate to insulin ratios, correction factors, and basal insulin doses, all explained with context for clinicians. Results from the study indicate that the TWIN system greatly improves the personalization of diabetes management by maintaining blood glucose levels within the target range for longer periods, reducing manual input and automating the critical aspects of diabetes care. However, the system's dependence on data quality from devices and its adaptability to a broader patient population need further development to ensure scalability and reliability. The TWIN system offers an innovative solution for pediatric diabetes management, using AI-driven digital twins to provide real-time, personalized recommendations that lighten the burden on patients and caregivers.

Another digital twin system introduced by Shamanna, P. et al. which investigated the use of digital twin technology enabled precision nutrition (TPN) for the management of Type 2 diabetes (T2D), focusing on reducing hemoglobin A1c (HbA1c) levels and minimizing the reliance on diabetes medications [27]. The researchers conducted a 90-day retrospective analysis involving patients with T2D who participated in the TPN Program. The system integrates CGM device data, food intake logs, and biometric data to provide personalized nutritional guidance, targeting postprandial glycemic responses (PPGR) [52]. Digital twins utilized to create virtual models of each patient's metabolic processes, enabling individualized nutrition recommendations. The system's machine learning model continuously refines its predictions based on real-time glucose readings and dietary habits, allowing the digital twin to deliver increasingly accurate advice on dietary Choices that minimize blood glucose spikes. After 90 days, patients in the TPN Program experienced notable improvements: HbA1c levels decreased from 8.8% to 6.9%, and insulin resistance decreased by 56.9%. Many patients were able to reduce or discontinue their diabetes medications, highlighting the program's effectiveness in improving glycemic control through precision nutrition. However, the reliance on patient reported food intake data introduces potential inaccuracies that could affect the precision of the digital twin's recommendations. Still, this application shows significant potential in managing T2D by providing tailored nutritional guidance and reducing medication dependency. However, further research with larger sample sizes and longer follow-up periods is

probably required to validate these findings and assess the long term sustainability of the improvements.

Digital twin applications for personalized medicine in mental health

Following our review of digital twin applications in diabetes care, in this section we turn to how this technology is helping in mental and neurological health. In mental health, digital twins create dynamic, data-rich models that capture each patient's unique mental and emotional profile. By integrating real-time data from behavioral patterns, cognitive markers, and physiological signals, these systems empower clinicians to anticipate challenges and design interventions adopted to each patient's specific needs.

Figure 6 presents a clear view of how digital twin models applied across mental health care, drawing on data from wearables, clinical assessments, and voice analytics. In the studies that follow, we examine their impact on migraines, dementia, Parkinson's, and depression, revealing how digital twins unlock new possibilities for real-time, individualized care.

Gazerani [53] Migraines are a recurring neurological disorder characterized by intense headaches and sensory disturbances, influenced by genetic, environmental, and lifestyle factors, making treatment complex and highly individualized [54]. Gazerani et al., in their study, proposed the use of intelligent digital twins for personalized migraine care. These digital twins create a virtual model of the patient, allowing for dynamic simulations that reflect individual symptoms, triggers, and responses to treatment. Key biomarkers, such as miRNAs (e.g., miR-34a, miR-382, miR-155) [55], have emerged as critical indicators for diagnosing and managing migraines. The authors proposed a

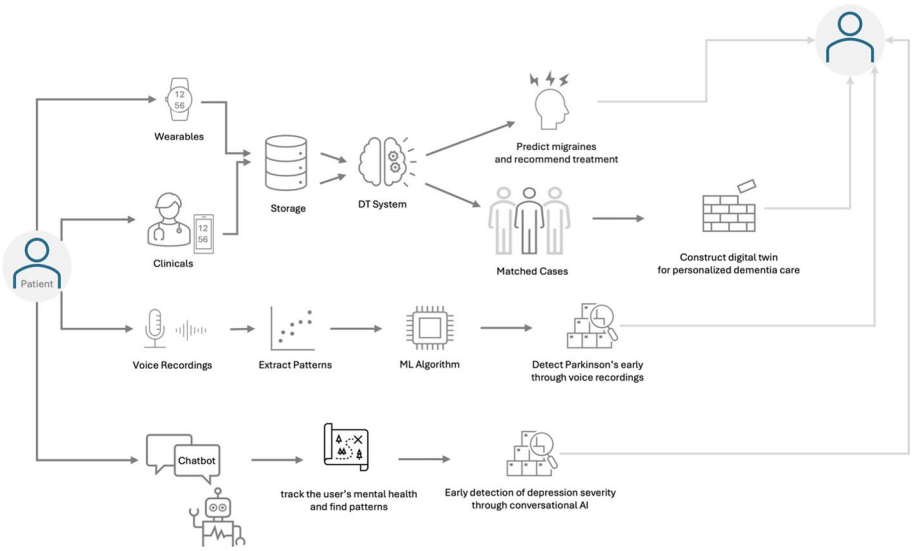


Fig. 6 Visualizing digital twin applications in mental and neurological healthcare. The diagram illustrates how various data sources, including wearable devices, clinical inputs, and voice recordings, contribute to digital twin systems, enabling personalised care. Applications include predicting migraines and adjusting treatments, constructing personalized digital twins for dementia care based on matched cases, detecting Parkinson's disease through voice analysis, and evaluating depression severity via conversational AI. These systems, powered by machine learning algorithms, enable personalized and proactive healthcare interventions

framework where the developed digital twin can leverage machine learning algorithms to analyze real-time data from wearable devices like the Empatica E4, which tracks biosignals such as heart rate, skin temperature, and electrodermal activity. This data is integrated with clinical information from the REFORM (Registry for Migraine Study) [56], which gathers biological markers, brain imaging, and treatment outcomes to identify predictive biomarkers, particularly for treatments like erenumab [57]. By refining the understanding of each patient's unique migraine triggers, the digital twin helps clinicians predict the onset of migraine attacks and adjust treatment strategies. The system simulates physiological responses, enabling personalized forecasting of symptoms and safer treatment adjustments, including simulations of drug interactions to reduce the risk of medication overuse.

Continuing from the work done in migraine care, another promising area for digital twin technology lies in dementia management. Dementia, a progressive neurological disorder marked by cognitive decline and memory loss, presents unique challenges in treatment due to the wide variability in symptom progression and patient responses [58]. Wickramasinghe et al. in their proposed Perspective, a digital twin application as a decision support system (DSS) for dementia care, accessible via mobile, tablet, or desktop [59]. Clinicians or carers input patient data (e.g., risk factors, symptoms, treatments) into the DSS, which connects to a remote server running a machine learning or deep learning algorithm. The algorithm retrieves similar cases from a dementia data repository containing rich data from an extensive cohort of dementia cases and constructs a digital twin by combining the best-matching cases. If no exact match is found, a similarity score is provided. This digital twin enables clinicians to make informed and personalized decisions regarding diagnosis and treatment planning. Since this is a perspective piece, the authors have not explored specific AI models or datasets that might underpin the development of a digital twin application. Although it doesn't showcase a real-world implementation, we included it as an insightful conceptual framework, offering a foundational approach for designing digital twin applications in healthcare with promising implications for personalized dementia care.

Expanding the potential of digital twins beyond dementia care, another study by Abirami et al. explores the use of this technology for early detection of Parkinson's disease [28]. The system constructs a patient-specific digital twin by analyzing voice recordings collected via smartphones or other devices. These recordings track vocal biomarkers associated with Parkinson's, such as vocal tremors and speech impairments, which are indicators of the disease's progression. The study utilized two datasets: DS1, consisting of voice data from 195 subjects (147 with Parkinson's, 48 healthy controls) collected from Kaggle Cerrahpasa Faculty of Medicine, Istanbul University repository [60], and DS2, from Oxford University [61]. The voice recordings were gathered using standard protocols to assess Parkinson's-related vocal impairments. Important vocal features were extracted from the recordings and processed by the core model, an Optimized Fuzzy-Based k-Nearest Neighbour (OF-kNN) classifier. This model was specifically chosen to handle the variability in voice data and offer more precise classifications of early Parkinson's symptoms. The OF-kNN classifier uses fuzzy logic to manage uncertainties in vocal feature measurements, enabling more refined detection of subtle changes in vocal biomarkers. This digital twin system allows clinicians to identify early signs of Parkinson's

disease and tailor treatments based on individual patient profiles. Its reliance on voice data makes it highly accessible and scalable for remote, continuous monitoring. The integration of the OF-kNN model within the digital twin framework highlights a forward thinking approach to personalized care, offering a more accurate and proactive method for early diagnosis and intervention in neurodegenerative diseases like Parkinson's.

Following the application of digital twins for early Parkinson's detection, a different approach extends the technology into the area of mental health. Abilkaiyrkyzy, A. et al. In their study, digital twins are adapted to assess depression through conversational AI [29]. With the help of natural language processing (NLP) techniques, this system creates a dynamic model of the user's mental state, engaging them in real-time dialogue to detect early signs of mental illness. Using a fine-tuned bidirectional encoder representation from transformers (BERT) [62] model trained on the Extended Distress Analysis Interview Corpus (E-DAIC) [63], which contains clinical interview transcripts, the system processes user responses, identifying the severity of depression from the conversation. The chatbot categorizes responses into different levels of depression severity, such as no symptoms, mild, or moderate/severe, providing personalized mental health insights. As users interact, their inputs help build a "digital twin" of their mental health, continuously refining the system's ability to predict their condition based on ongoing conversations.

The system was tested on the E-DAIC dataset and real-world participants, achieving an accuracy of 69% and demonstrating high usability. This system represents a significant advancement in AI-driven mental health screening, enabling early detection through everyday dialogue.

As we have seen, digital twins are helping to address various mental health disorders, particularly conditions like depression and neurodegenerative diseases. Yet, challenges remain, including safeguarding data privacy, ensuring quality input, and smoothly integrating these systems into healthcare. Despite these hurdles, digital twins hold remarkable potential to expand care options across mental health and chronic diseases. Machine learning drives this progress, with its ability to analyze complex data and produce targeted insights. We expect a high impact of digital twins in the coming future, relying on continued advancements in machine learning.

Different practical implementations of digital twins across healthcare domains

Moving beyond the specialized focus areas like diabetes, cardiac care, and mental health, digital twins have recently expanded into a wider range of healthcare applications. These developments overcome practical, everyday health challenges, offering new ways to improve patient outcomes. Figure 7 provides a visual summary of different digital twin applications within healthcare, showing the distinct processes and outcomes for each use case across different medical challenges.

Sarp et al. developed a digital twin framework designed specifically for chronic wound management, addressing a critical need for monitoring and personalized treatment in wound care [30]. Chronic wounds, often slow to heal and challenging to manage, require continuous assessment and adjustment of treatment strategies [64]. This study introduces an innovative approach that allows clinicians to visualize and predict the healing process of chronic wounds using digital twin technology.

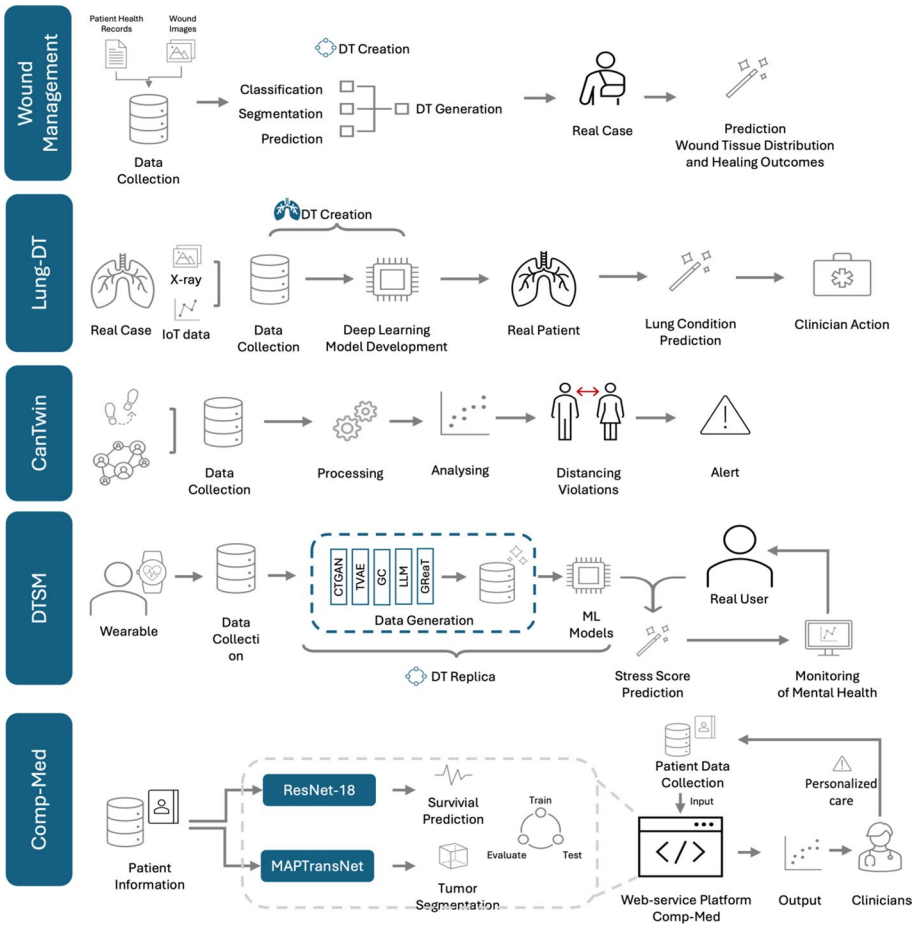


Fig. 7 Visual representation of diverse digital twin applications across the healthcare domain. This diagram outlines the processes from data collection to real-time prediction in various areas: wound management, lung health monitoring (Lung-DT), social distancing management (CanTwin), and stress monitoring. Each workflow showcases the integration of machine learning models and real-time data to enable predictive, personalized healthcare interventions

The developed framework uses machine learning models to simulate the progression of wound healing. By creating a virtual replica of the patient's wound, the system enables healthcare providers to monitor changes, assess the effectiveness of treatments, and anticipate potential complications. The key advantage lies in its ability to offer personalized insights into the healing process, allowing clinicians to make data-driven decisions on whether to continue or adjust a treatment plan. To achieve this, the authors employed a generative adversarial network (GAN) [65] as the core model. GANs are well-suited for generating realistic future outcomes, making them ideal for simulating wound Healing trajectories. The model was trained on a dataset of 700 paired wound images, capturing the before and after conditions of various chronic wounds over time. This model not only predicts the wound's future state but also simulates potential healing outcomes based on different treatment scenarios. The model operates in three stages: first, classifying the wound type; next, segmenting and classifying the wound tissues; and finally, predicting the distribution of wound tissues over time. Testing involved comparing the predicted tissue distribution to actual outcomes, with the model achieving significant

accuracy in identifying non-healing wounds and predicting the overall healing process. The data used includes a combination of wound images and patient demographic information such as age, underlying conditions, and the wound's medical history. This holistic data approach allows the model to consider multiple variables influencing wound healing, making the predictions more robust and personalized. The integration of image data with patient characteristics enhances the framework's capacity to adapt to individual patient needs. The system's performance was evaluated using the mean squared error (MSE) metric, which compared predicted tissue distributions to actual results. The MSE score of 25.84 indicated that the model accurately predicted around 74% of the wound tissue.

This system enables remote wound monitoring and early detection of complications, improving outcomes and reducing costs. While challenges like data privacy and broader validation exist, this AI-powered approach offers real-time, personalized insights that enhance patient care and treatment strategies.

Moving on to another area of respiratory health, Avanzato et al. developed Lung-DT, a framework specifically focused on improving lung disease diagnosis and real-time monitoring. It's an AI-powered digital twin framework for thoracic health monitoring and diagnosis, aimed at improving the diagnosis and management of lung diseases, particularly in the context of COVID-19 [66] and other lung-related pathologies [31]. The framework seeks to address the growing need for timely, accurate diagnosis and continuous monitoring of lung health through real-time data integration and AI-driven analysis.

The developed application provides a digital twin of the respiratory system, enabling healthcare providers to monitor and assess lung conditions remotely. This virtual representation integrates chest X-ray images and real-time physiological data collected from IoT sensors, such as blood oxygen levels (SpO2), to generate predictive insights about the patient's lung health. The system can classify lung conditions into five distinct categories: normal, COVID-19, lung opacity, pneumonia, and tuberculosis, allowing clinicians to make informed decisions based on accurate, up-to-date data.

The proposed Lung-DT model relies on the YOLOv8 deep learning model [67], which is highly effective for real-time image detection and classification tasks. YOLOv8 was specifically trained to detect anomalies in chest X-rays, such as opacities associated with COVID-19 or pneumonia. The model was trained using a large dataset of chest X-ray images, spanning different lung conditions, and enriched with IoT sensor data to create a more comprehensive view of the patient's health. The integration of SpO2 data alongside the image analysis allows for a more precise diagnostic tool, ensuring that the predictions are grounded in multiple physiological indicators rather than just visual analysis. The dataset used for model training consisted of thousands of labeled chest X-ray images from public datasets, covering various lung conditions such as COVID-19, pneumonia, and tuberculosis. Additionally, real-time physiological data from IoT sensors, like SpO2 measurements, were incorporated to provide a dynamic and real-time monitoring capability. This combination of static and dynamic data allows the system to simulate real-world scenarios and predict disease progression more accurately. The evaluation of the Lung-DT framework showed high performance in classifying lung conditions. The YOLOv8 model achieved an accuracy of 93.4% in detecting anomalies in chest X-rays

across the five categories. In addition, the integration of physiological data from IoT sensors further enhanced the accuracy of the system, ensuring more reliable diagnostics by considering multiple health indicators.

The Lung-DT framework enables early detection and remote monitoring of lung diseases, minimizing hospital visits and allowing timely interventions. By tracking physiological data and using predictive insights, it helps prevent severe health declines. Though challenges like model adaptability and data privacy remain, this AI-powered tool shows strong potential for personalized lung care.

Another important application, also focused on lung health, is the precision medicine platform proposed by Kolekar et al., which leverages AI for personalized care strategies [32]. The web-services platform (Comp-Med) is designed to support clinicians in delivering highly personalized treatments by integrating AI-powered services with patient data from multiple sources. It focuses on three core areas: predicting lung cancer survival, analyzing lung Tumor segmentation for survival analysis, and providing rapid response care in emergencies. CompMed's predictive power lies in two key AI models. For lung cancer survival prediction, the system employs ResNet-18 [68], a deep learning model that performs binary classification on five-year survival rates. The integration of multi-modal data, including positron emission tomography (PET) and computed tomography (CT) scans, enables this model to generate highly accurate predictions. Additionally, the platform incorporates MAPTransNet [69], a global context-aware transformer model used for lung Tumor segmentation. MAP-TransNet plays a crucial role in identifying Tumor regions on PET images, aiding in survival analysis and treatment planning. The datasets used to train and validate these models are substantial. For the lung cancer prediction model, the study utilized the LC-CNUHH dataset from Chonnam National University Hwasun Hospital, which includes 4,591 lung cancer cases, encompassing both non-small cell lung cancer (NSCLC) and small cell lung cancer (SCLC) patients. For the rapid response analysis, the researchers used the RRT-CNUHH dataset, containing data from 25,329 patients, focusing on vital signs to predict clinical deterioration in emergency settings. Model performance was evaluated using various metrics. The lung cancer prediction model achieved an impressive C-index of 0.97 for five-year survival classification. Tumor segmentation accuracy was measured using the Dice score, while survival analysis relied on metrics such as the Brier score, Mean Absolute Error (MAE) [70], and Binomial Log-likelihood (IBLL). The rapid response model for emergency care demonstrated strong performance with an area under the Receiver Operating Characteristic curve (AUROC) [71] of 0.837, highlighting its ability to accurately predict patient deterioration in real-time.

CanTwin system, developed by De Benedictis et al. in collaboration with Hitachi [72]. This system was designed for a canteen serving 1100 workers to monitor social distancing, manage queues, count people, and supervise table occupancy during the COVID-19 pandemic. The study emphasizes the growing role of digital twins in public health to enhance safety and space management in industrial settings.

The CanTwin architecture integrates several layers, starting with the physical twin, which mirrors the actual canteen environment, and the data collection layer, which gathers real-time data from sensors and IoT devices. This information is processed and used

to update the digital twin model in real time. An analytics layer applies algorithms to detect social distancing breaches and predict crowding, while the user interface layer presents these insights through dashboards, enabling managers to take quick action.

Data for the system was gathered from sensors and IoT devices tracking foot traffic and occupancy. This data was used to train and assess the system's accuracy in detecting social distancing breaches, predicting congestion, and providing real-time alerts. Although specific quantitative results aren't given, the paper emphasizes the system's effective deployment in managing crowd control and optimizing space usage in a real-world environment.

Digital twin systems have also supported individuals facing stress-related challenges, helping to improve their quality of life. Kumi et al. proposed a digital twin framework for stress management by utilizing synthetic data to predict stress management scores based on data collected from wearables like the Google Pixel watch [33]. This system, dubbed Digital Twin for Stress Management (DTSM), aims to address the challenges of limited data and privacy concerns by generating high quality synthetic data to simulate real-world conditions.

The DTSM integrates multiple synthetic data generation techniques, including Conditional Tabular GAN (CTGAN) [73], Tabular Variational Autoencoder (TVAE) [73], Gaussian Copula [74], and two Large Language Model based approaches: Realistic Relational and Tabular Transformer (REaLTabFormer) [75] and Generation of Realistic Tabular data (GReaT) [76]. Among these, REaLTabFormer demonstrated the highest performance in replicating real-world data distributions, achieving a data quality score of 93%.

The data used to build the DTSM includes heart rate variability (HRV), sleep patterns, and physical activity metrics collected from two subjects wearing the Google Pixel watch. Features such as resting heart rate, deep sleep, rapid eye movement (rem) sleep, light sleep, and daily step counts and much more were extracted from the wearables, and the synthetic data module modeled this dataset to generate digital twins for stress score prediction.

After generating the synthetic data, the authors trained four machine learning models, XGBoost, Random Forest, Light Gradient Boosting Machine (LightGBM), and TabNet on the synthetic data to predict stress management scores. The LightGBM model, trained on REaLTabFormer's synthetic data, performed the best, achieving an MAE [70] of 3.39%, MAPE of 4.60%, and an R^2 of 0.63, demonstrating its efficacy for this task.

The results show DTSM's potential for effective stress monitoring and prediction through synthetic data, providing a scalable, privacy-conscious solution for stress management in healthcare.

In this section, we explored different applications spanning multiple healthcare domains, it became clear that this technology's adaptability extends beyond traditional clinical settings. While these systems offer significant potential for personalized and predictive healthcare, challenges such as model generalization, data privacy, and transparency remain. Addressing these gaps will be essential for realizing the full impact of digital twins in diverse healthcare applications.

Integrating LLM models and graph theory in digital twins applications

The rise of Large Language Models (LLMs) is revolutionizing digital twin applications, especially in healthcare [35, 77]. These models address complex challenges in patient care and research, with refined chatbots enhancing clinician interaction [78]. In the next section, we explore three studies that integrate LLMs with unique methodologies for healthcare, as illustrated in Fig. 8, showcasing each study's approach and impact.

TWIN-GPT [34], developed by Wang et al., addresses the critical problem of clinical trials being hampered by limited data and the complexity of individual patient profiles. The proposed model leverages large language models to create personalized digital twins for patients participating in clinical trials. These digital twins allow for more accurate prediction of clinical trial outcomes, even in the face of limited EHR data [79]. TWIN-GPT works by predicting patient responses to treatments based on their real-time and historical health data. The model fine-tunes a pre-trained LLM model (ChatGPT [80]), using clinical trial datasets, generating a virtual twin of each patient that mimics their health trajectory and response to interventions. TWIN-GPT can also simulate counterfactual scenarios, predicting what would happen under different treatment paths. This capability enhances the precision of clinical trial design, potentially reducing costs and improving safety by allowing for virtual testing before real-world application. The dataset

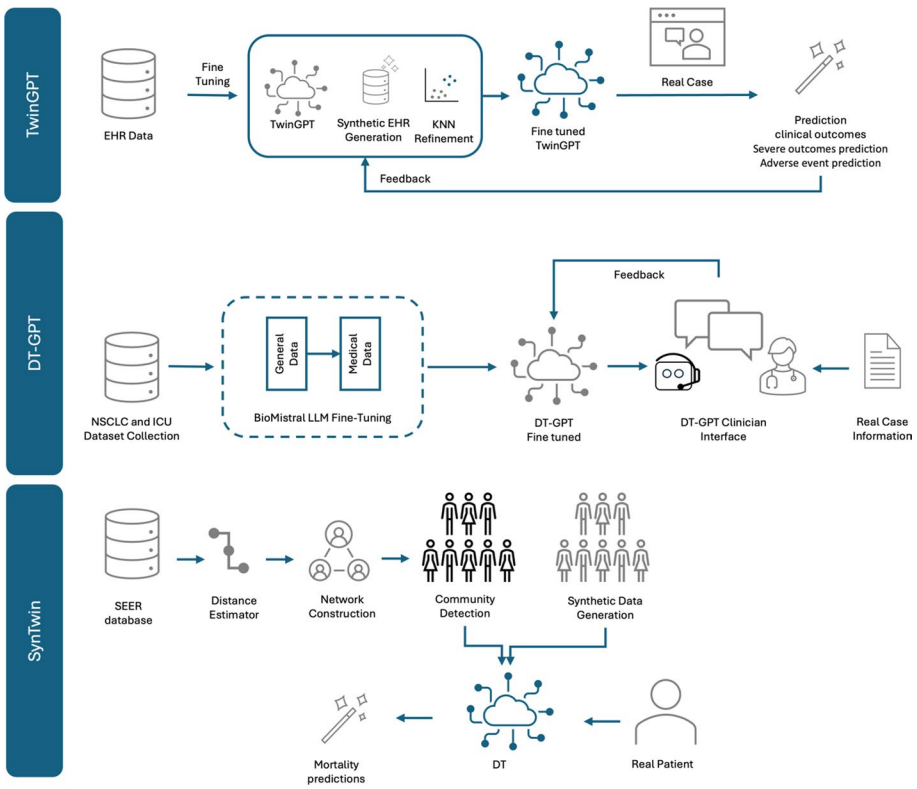


Fig. 8 Visual representation of the workflows for three advanced digital twin applications TwinGPT, DT-GPT, and SynTwin, leveraging large language models and graph theory in healthcare. TwinGPT focuses on refining electronic health records (EHR) through synthetic data generation and k-nearest neighbors (KNN) refinement. DT-GPT fine-tunes biomedical LLMs to forecast clinical outcomes from NSCLC and ICU patient datasets. SynTwin uses a network-based approach to predict patient mortality in oncology by constructing a synthetic patient population from the SEER dataset

used for model training includes a Phase III breast cancer trial dataset (NCT00174655) consisting of 2887 patients, along with the multi-modal Trial Outcome Prediction (TOP) dataset, which comprises over 9000 clinical trials from Phase I to Phase III. The dataset integrates drug molecule information, disease codes, and trial outcomes, providing a robust platform for training the LLM. In terms of evaluation, TWIN-GPT outperforms baseline models such as EVA [81], SynTEG [82], PromptEHR [83], and TWIN-VAE [84] in several key metrics. The model achieves a high AUROC [71] score of 0.821 in predicting severe outcomes and 0.838 when using real data. For adverse event prediction, it performs similarly to real data, with AUROC [71] scores close to 1. These results validate the model's ability to generate high-fidelity synthetic data, which closely mirrors real clinical trial data. Moreover, As privacy is a critical concern in clinical trials, the authors mentioned that privacy protection metrics such as presence and attribute disclosure were significantly lower for TWIN-GPT compared to other methods, ensuring patient confidentiality and providing a secure framework for virtual clinical trials while ensuring accurate patient outcome predictions.

As TWIN-GPT showcases the potential of LLM-driven digital twins in refining clinical trials, DT-GPT further extends this approach into real-time clinical care by forecasting patient health trajectories [35]. Developed by Marakov et al., DT-GPT is a digital twin model designed to enhance clinical outcome forecasting by leveraging advanced LLMs. This model addresses a significant challenge in clinical care for accurately predicting patient health trajectories, which is often complicated by the high variability and incomplete nature of clinical data, particularly in complex cases such as non-small cell lung cancer (NSCLC) and intensive care unit (ICU) patients. Traditional approaches frequently face limitations due to the need for data imputation and normalization, and they often struggle to capture the intricate relationships among patient metrics. DT-GPT, however, takes a different approach by utilizing the robustness of LLMs to process raw, real-world patient data with minimal adjustments, effectively sidestepping these limitations.

DT-GPT's architecture is anchored in BioMistral [85], a biomedical LLM fine-tuned specifically to enhance the model's medical context awareness. This foundation enables DT-GPT to operate without the need for imputation or normalization, making it particularly adept at handling fragmented and noisy clinical records. The model preserves the intricate relationships between clinical variables, utilizing cross-variable dependencies to strengthen its forecasting abilities. This capability allows DT-GPT to adaptively learn patterns in patient data, predicting health metrics and outcomes over both short and long-term periods.

The study utilises two substantial datasets: the first comprises long-term data from 16,496 NSCLC patients (773,607 patient-days) sourced from the Flatiron Health database, while the second dataset includes 35,131 ICU patients with 1,686,288 time points from the MIMIC-IV database. For evaluation, the authors benchmark DT-GPT against established models such as LightGBM. The results are notable, with DT-GPT achieving a 3.4% improvement in mean absolute error (MAE) [70] on the NSCLC dataset and a 1.3% improvement on the ICU dataset. Additionally, DT-GPT maintains high R^2 scores (0.98 for NSCLC and 0.99 for ICU), underscoring its ability to preserve the variable relationships integral to clinical data.

A distinctive feature of DT-GPT is its zero-shot prediction capacity. This aspect allows the model to outperform fully trained models even on variables it has not previously encountered, a significant advantage in adapting to diverse clinical scenarios. Evaluations, including AUROC [71] and Mean Absolute Error (MAE) [70], underscore DT-GPT's robust performance across time, and the authors highlight an interactive explainability feature, where clinicians can engage with the model via a chatbot interface. This capability provides valuable insights into the key variables influencing patient outcomes, reinforcing DT-GPT's role as a practical tool for personalized, predictive healthcare.

Moving on from applications of large language models to SynTwin [36], a graph-based digital twin framework developed by Moore et al., provides a digital twin application to predict patient mortality in oncology while preserving privacy. Using the SEER dataset with 10 million cancer cases, SynTwin generates synthetic profiles that mirror real patient complexity.

The process begins with calculating distances between patients using metrics like Euclidean [86], Manhattan [87], Cosine, and Gower [88], with Gower performing best due to its flexibility with both discrete and continuous data. A network of patient nodes is then created, filtering edges based on a percolation threshold to prevent a fully connected network. Using the Multilevel algorithm, the model detects patient communities, which serve as a foundation for generating synthetic patients. Synthetic data generation employs three algorithms: categorical latent Gaussian process (CLGP) [89], mixture of product of multinomials (MPoM) [90], and multi-categorical extension of a medical generative adversarial network (MC-MedGAN) [91], each chosen for their ability to replicate the statistical characteristics of the original data. Digital twins are selected from these synthetic patients, ensuring they align closely with real patient profiles.

For mortality prediction, the model uses a nearest-neighbor approach within each community, simulating patient-specific outcomes with improved predictive accuracy. SynTwin achieved an AUROC [71] of 0.864 (95% CI: 0.857–0.872), significantly outperforming predictions made using real data alone (AUROC [71] of 0.791, 95% CI: 0.781–0.800). These results underscore the framework's value in precision medicine, where privacy-conscious synthetic data solutions can enhance predictive accuracy and patient outcome forecasting.

In closing this section, the studies we reviewed outline the practical impact of these technologies, demonstrating their potential to enhance our quality of life in meaningful ways.

Ethical, regulatory, and implementation considerations

The integration of digital twins (DTs) into healthcare systems offers exciting opportunities for more personalized, predictive, and responsive care, but it also brings to the surface a set of complex and interrelated challenges [53] that must be addressed before these technologies can be used responsibly in clinical settings. One of the most pressing concerns is data *privacy and security* [29].

DTs rely heavily on real-time streams of sensitive patient data from wearables, mobile devices, and electronic health records, and managing this data safely becomes increasingly difficult as the volume and complexity of inputs grow. In many cases, especially in

mental health applications, the risk of misinterpretation or misuse is heightened due to the nuanced and deeply personal nature of the data.

Addressing these risks requires a combination of technical safeguards and ethical practices. Anonymization and de-identification techniques are essential to ensure personal identifiers are removed from datasets, and the use of synthetic data, artificially generated datasets that mimic real ones can help train and test models without exposing actual patient information. Secure data transfer protocols, encrypted storage systems, and access controls that give patients agency over their data use are also critical [72]. At the infrastructure level, edge-based systems that process data locally can reduce exposure by limiting transmission over networks, while explicit consent mechanisms and limited retention periods help reinforce trust and transparency [26, 31].

Alongside these privacy measures, *regulatory considerations* also demand attention. Many of the current approval processes from agencies like the Food and Drug Administration (FDA) and European Medicines Agency (EMA) are not yet well-suited to the evolving nature of DTs, which continuously learn and adapt over time [72]. Traditional validation frameworks assume static systems, whereas DTs require ongoing monitoring and re-evaluation [53]. This creates uncertainty around how safety and effectiveness should be measured, and when a model is considered ready for clinical use. The absence of standardized validation protocols, limited availability of high-quality and diverse datasets, and a lack of clear guidelines for measuring predictive uncertainty all make regulatory approval more difficult [31]. Ethical review processes and compliance with institutional review boards remain essential, especially when working with real patient data.

In parallel, *clinician adoption* poses another major challenge. Even the most advanced DT system cannot be effective if clinicians are hesitant to use it. A common barrier is the lack of transparency in how model outputs are generated. Many AI-based systems function as "black boxes", offering predictions or recommendations without explaining how they were reached [30]. For clinicians who are trained to reason through decisions, this lack of clarity can reduce confidence and make integration into care processes more difficult [31]. Moreover, DTs must fit within existing clinical workflows, which vary across institutions and often rely on legacy systems. Integrating DTs with electronic health record (EHR) platforms, ensuring stable network performance, and preparing institutions for the technical demands of DT tools all require time, planning, and investment [35]. In addition, many healthcare professionals have limited training in interpreting AI-driven tools, and ongoing interdisciplinary collaboration between clinicians and developers is necessary to ensure usability.

To support meaningful adoption, several practical steps can be taken: design DT systems with explainability in mind so that clinicians can understand and trust the results; align DT development with regulatory expectations from the early stages; embed DTs into the digital infrastructure clinicians already use, such as EHR platforms; and conduct real-world pilot studies in clinical environments to evaluate effectiveness, usability, and trustworthiness in practice. These efforts will not only help DTs meet technical standards, but also prepare the healthcare ecosystem for their responsible and sustainable use.

Conclusion

This review examined how digital twin (DT) technologies are being applied across healthcare domains, with particular attention to the role of artificial intelligence in supporting these systems. These technologies reveal transformative potential from diabetes care to cardiovascular disease management and mental health monitoring, where they already contribute to more personalized, dynamic, and anticipatory care models by synthesizing patient-specific data, simulating outcomes, and informing clinical decisions in real time. These advancements are evidenced by the shift of DTs from theoretical concepts to real-world implementations integrated into clinical workflows. Even with this progress, these promising developments coexist with persistent limitations. Many models remain constrained by their reliance on structured, high-quality datasets that missing universal accessibility or standardization, forcing custom solutions for each new setting and exposing critical gaps in scalability. A disconnect remains between technical capability and practical implementation, where institutional support, clinician engagement, and operational alignment often struggle to keep pace with technological progress. On top of these challenges, there are still important concerns about system safety and transparency, which grow more urgent as DTs increasingly guide clinical judgment. Strong safeguards are needed for output validation, failure mode identification, and interpretability, necessitating not only routine benchmarking against clinical standards but also clear protocols for handling messy data and defining accountability boundaries between algorithms and practitioners. At the same time, data security also remain a serious issue. The essential fuel of digital twins high resolution, real-time patient data streams introduces unprecedented vulnerabilities that threaten adoption unless countered by rigorous governance, encryption, access controls, and ethical synthetic data practices capable of preserving clinical trust. Another key feature of digital twin systems is their dynamic architecture, unlike traditional models, a DT is designed to evolve as new information becomes available adjusting predictions, refining simulations, and modifying recommendations in response to patient changes. While this feedback loop is a core strength, it also introduces added complexity. Systems must be built to manage continuous updates, track versioning, and ensure that iterative learning does not compromise model consistency or reproducibility. Reaching the full potential of DTs in healthcare will depend on a careful balance between technical innovation and implementation discipline. Efforts to improve model generalizability, explainability, and adaptability must go hand in hand with safeguards that ensure patient safety, data protection, and ethical use. Cross-disciplinary collaboration spanning clinical practice, engineering, data science, and health policy will be central to advancing these technologies responsibly. The evidence leaves no doubt, while technical foundations continue their rapid evolution, parallel progress in ethical and operational frameworks must accelerate to match their growth an imperative this review crystallizes through its comprehensive analysis of both achieved milestones and the critical challenges that will define the next era of healthcare innovation.

Abbreviations

AI	Artificial intelligence
AS	Aortic stenosis
AUPRC	Area under the precision-recall curve
AUROC	Area under the receiver operating characteristic curve
BERT	Bidirectional encoder representations from transformers

CGM	Continuous glucose monitoring
CLGP	Categorical latent Gaussian process
CPPS	Cyber-physical production systems
CT	Computed tomography
CTGAN	Conditional tabular generative adversarial network
CVD	Cardiovascular disease
DICE	Dice similarity coefficient
DSS	Decision support system
DT	Digital twin
DTSM	Digital twin for stress management
E-DAIC	Extended distress analysis interview corpus
ECG	Electrocardiogram
EHR	Electronic health records
FHR	Fetal heart rate
FHRV	Fetal heart rate variability
GAN	Generative adversarial network
GPT	Generative pre-trained transformer
HbA1c	Hemoglobin A1c
HRV	Heart rate variability
IBLL	Binomial log-likelihood
ICU	Intensive Care Unit
IoU	Intersection over union
IoT	Internet of things
LCSA	Lung cancer survival analysis
LLM	Large language model
LR	Logistic regression
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MC-MedGAN	Multi-categorical medical generative adversarial network
MDI	Multiple daily insulin injections
MIMIC-IV	Medical information mart for intensive care IV
miRNA	MicroRNA
MPoM	Mixture of product of multinomials
MSE	Mean squared error
NASA	National Aeronautics and Space Administration
NCD	Non-communicable disease
NLP	Natural language processing
NSCLC	Non-small cell lung cancer
OF-kNN	Optimized fuzzy-based k-nearest neighbor
PET	Positron emission tomography
PPGR	Postprandial glycemic response
QC	Quality control
REaLTabFormer	Realistic relational and tabular transformer
REFORM	Registry for migraine study
RMSE	Root mean square error
SCLC	Small cell lung cancer
TAVR	Transcatheter aortic valve replacement
T1D	Type 1 diabetes
T2D	Type 2 diabetes
TVAE	Tabular variational autoencoder
TWIN-GPT	Digital twin generative pre-trained transformer

Supplementary Information

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Additional file 1

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Conceptualization, ZG, MTA and RA; analysis, ZG, MTA and RA; methodology, ZG, MTA and RA.; project administration, RA.; supervision RA; validation, ZG, MTA and RA; writing–original draft, ZG and MTA; writing—review and editing, ZG, MTA and RA. All authors have read and agreed to the published version of the manuscript.

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Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate
Not applicable. This study is a review of existing literature and does not involve any human participants, human data, human tissue, or animal subjects.

Consent for publication
This study is a literature-based review and does not include any individual person's data in any form.

Competing interests
The authors declare no competing interests.

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